

0, 4, 2, 3, 4 $\Rightarrow$ 2 hours

Lec 5 $\Rightarrow$ 3-4 videos

Practice Questions on time Complexity

Telugoon group:- description

$$\boxed{n = 12000}$$

Time complexity?

✓ 10) Main()

{ i=10

while(i<=n)

{

i=i*12

}

}

$$i = 10,$$

$$i <= n$$

$$i = i * 12$$

$$i = 10,$$

$$10 < 1200,$$

$$i = 10 \times 12$$

$$i = 120$$

$$i = 10 \times 12,$$

$$120 < 1200,$$

$$i = 120 \times 12$$

$$i = 120 \times 12$$

$$i = 12 \times 12 \times 10 \quad 1440 < 12000$$

$$i = 1440$$

$$i = 12 \times 12 \times 12 \times 10 \geq \underline{120000}$$

$$\boxed{i = 12^x \times 10} \geq n$$

$$\log_a(m \times n) = \log_a m + \log_a n$$

$$I = 12^n \times 10$$

$$12^n \times 10 = n$$

side taking log, on base 12

$$\log_{12}(12^n \times 10) = \log_{12} n$$

$$\Rightarrow \log_{12} 12^n + \log_{12} 10 = \log_{12} n$$

$$\Rightarrow n \log_{12} 12 = \log_{12} n - \log_{12} 10$$

$$\Rightarrow n = \log_{12} n - \log_{12} 10$$

$$O(\log_{12} n) \quad O(\log_{12} n)$$

n-1
n-2
n-100
n+100

~~O(n)~~
Both

$\boxed{x}$ $12 \rightarrow 36 \rightarrow 72$

$l = 1$, $l = 1 \times 12$
 $l = 12 \times 3$
 $l = 36$

$n = 18000$

$l = 72$

11.) Main()

{ $i=1$

while($i \leq n$)

{

$i = i * 12$

$i = i * 3$

$i = i * 2$

}

}

$l = l \times 72$

$l = 1$, $1 \leq 18000$,

$l = 72$, $72 \leq 18000$,

$l = 72 \times 72$

$l = 72^x \times 1$

$l = 1 \times 72$

$l = 72$

$l = 72 \times 72$

$$O(\log_{72} n)$$

$$\Rightarrow 72^x \approx n$$

$$\log_{72} 72^x = \log_{72} n$$

$$x \log_{72} 72 =$$

$$\log_{72} n$$

$$x =$$

$$10^{11}$$

while($x \leq n$)

$$x = x \times b$$

1

$$b^x \times a = n$$

$$10$$

$$12^x \times 10 = n$$

$$11$$

$$1 \times 72^x = n$$

12.) Main()

```
{
  for(i=1; i<=n; i++)
  {
    for(j=1; j<=n; j++)
    {
      x=y+z
    }
  }
}
```

$O(n^2)$

$O(n^2)$
ans

$O(n^2)$

```
for(l=1; l<=n; l++)
{
  print('hi')
}
```

l=1, 1<=n ;
l=2, 2<=n ;
l=3, 3<=n ;
:
:
l=n, n<=n

n=4
hi, hi, hi, hi
times
l++
l=2
l=3
l=4
:
:
l=n+1

n+n+n+n - - -
n [1+1+1+1] = n^2 - - -
n times

$$n = \underline{10000}$$

11.) Main()

```
{  
  while(n >= 12)  
  {  
    n = n/7  
    n = n/2  
    n = n/5  
  }  
}
```

$$n = n/70$$

$$n = n/70$$

$$n = \frac{n}{70^2}$$

$$n = \frac{n}{70^3}$$

$$n \Rightarrow \frac{n}{70^k}$$

$$\frac{n}{70^k} < 12$$

$$n = n/70$$

$$n = \boxed{\frac{n}{70^L} \quad L=12}$$

$$(1) \Rightarrow n = n/70$$

$$(2) \Rightarrow n = (n/70)/70$$

$$n = n/70^2$$

$$(3) \Rightarrow n = (n/70^2)/70$$

$$(4) \Rightarrow n = n/70^3$$

$$n = 12 \times 70^L$$

$$\Rightarrow \log_{70} n = \log_{70} 12 + \log_{70} 70^L$$

$$\Rightarrow \log_{70} n = \log_{70} 12 + L \log_{70} 70$$

$$n = 70^{\log_{70} n}$$

$$O(\log_{70} n)$$

0, 1, 2, 4, 8, 16, 32

$$n = 1024$$

```

12.)main()
{
    i=2;
    while(i<=n)
    {
        i=i^2
    }
}

```

$$\begin{aligned}
 i &= 2^0 & i &= 2^1 \\
 i &= 2^1 & i &= 2^2 \\
 i &= 2^2 & i &= 2^4 \\
 i &= 2^4 & i &= 2^8 \\
 i &= 2^8 & i &= 2^{16}
 \end{aligned}$$

$$\begin{aligned}
 i &= 2, & 2 &\leq 1024, \\
 i &= 2^2, & 2^2 &\leq 1024, \\
 i &= (2^2)^2, & (2^2)^2 &\leq 1024, \\
 i &= ((2^2)^2)^2
 \end{aligned}$$

$$\begin{aligned}
 i &= 2^2 \\
 i &= (2^2)^2 \\
 i &= ((2^2)^2)^2
 \end{aligned}$$

$$\boxed{i = 2^{2^x}} \geq n$$

$$\begin{aligned}
 i &\geq n \\
 2^{2^x} &= n
 \end{aligned}$$

$$2^x = n$$

$$\log_2 2^x = n$$

$$x \log_2 2 = \log_2 n$$

$$x = \log_2 n$$

$$\log_2 2^x = \log_2 \log_2 n$$

$$\Rightarrow x \log_2 2 = \log_2 \log_2 n$$

$$x = \log_2 \log_2 n$$

$$O(\log_2 \log_2 n)$$

